

Foundation (Jalapeno)

- Q1.** What is the result of $(f+g)(x)$ if $f(x) = x^2$ and $g(x) = 2x$?
(A) $x^2 + 2x$
(B) $x^2 - 2x$
(C) $2x^2$
(D) $x^2 + 1$
(E) $2x^2 + 1$
- Q2.** If $f(x) = x - 1$ and $g(x) = x + 1$, what is $(f \cdot g)(x)$?
(A) $x^2 - 1$
(B) $x^2 + 1$
(C) x^2
(D) x
(E) $x - 1$
- Q3.** What is the domain of $(f/g)(x)$ if $f(x) = x^2$ and $g(x) = x$?
(A) $(-\infty, 0) \cup (0, \infty)$
(B) $(-\infty, 0)$
(C) $(-\infty, \infty)$
(D) $[0, \infty)$
(E) $(-\infty, -1] \cup [1, \infty)$
- Q4.** If $f(x) = 2x + 1$ and $g(x) = x - 2$, what is $(f - g)(x)$?
(A) $x + 3$
(B) $x - 3$
(C) $x + 1$
(D) $x - 1$
(E) $3x - 1$
- Q5.** What is the result of $(f/g)(x)$ if $f(x) = x^2 - 1$ and $g(x) = x + 1$?
(A) $x - 1$
(B) $x + 1$
(C) x^2
(D) x
(E) $x^2 - 2x + 1$
- Q6.** If $f(x) = x^2$ and $g(x) = 2x$, what is the domain of $(f + g)(x)$?
(A) $(-\infty, \infty)$
(B) $(-\infty, 0) \cup (0, \infty)$
(C) $[0, \infty)$
(D) $(-\infty, 0)$
(E) $(-1, 1)$
- Q7.** What is $(f \cdot g)(x)$ if $f(x) = x + 1$ and $g(x) = x - 1$?
(A) $x^2 - 1$
(B) $x^2 + 1$
(C) x^2
(D) x
(E) $x^2 - x$
- Q8.** If $f(x) = 2x$ and $g(x) = x - 2$, what is $(f + g)(x)$?
(A) $3x - 2$
(B) $3x + 2$
(C) x
(D) $x - 2$
(E) $2x + 1$

Open Ended Questions (FRQ)

- Q9.** If $f(x) = x^2 - 1$ and $g(x) = x + 2$, find $(f \cdot g)(x)$ and determine its domain.
- Q10.** Given $f(x) = x - 3$ and $g(x) = 2x + 1$, find $(f + g)(x)$, $(f - g)(x)$, and compare their domains.